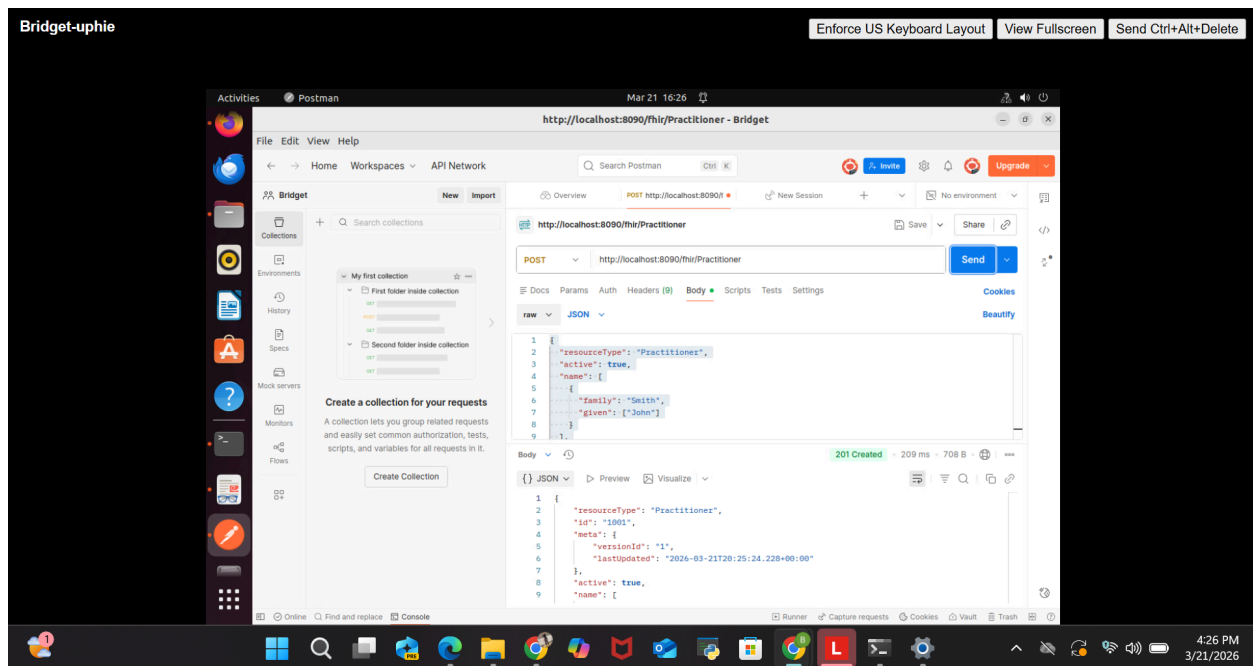


Architecture Development Part 5: Interoperability- FHIR Data Exchange with HAPI-FHIR

- Take a screenshot, save as response.code.png (or other picture format extension), and submit with your assignment on canvas demonstrating successful response code. (The screen short must show the name of your VM) (5 points)



1. What is the process to create a new FHIR resource using Postman and HAPI FHIR server? (5

points)

These are the steps below

i. First, launch Postman and log in using your credentials. Then create a new HTTP request and set the method to POST to send data to the server and create a new resource.

ii. Next, set the request URI to:

<http://localhost:8090/fhir/Practitioner>

where <http://localhost:8090/fhir> is the base URL and */Practitioner* is the endpoint for creating a new Practitioner resource.

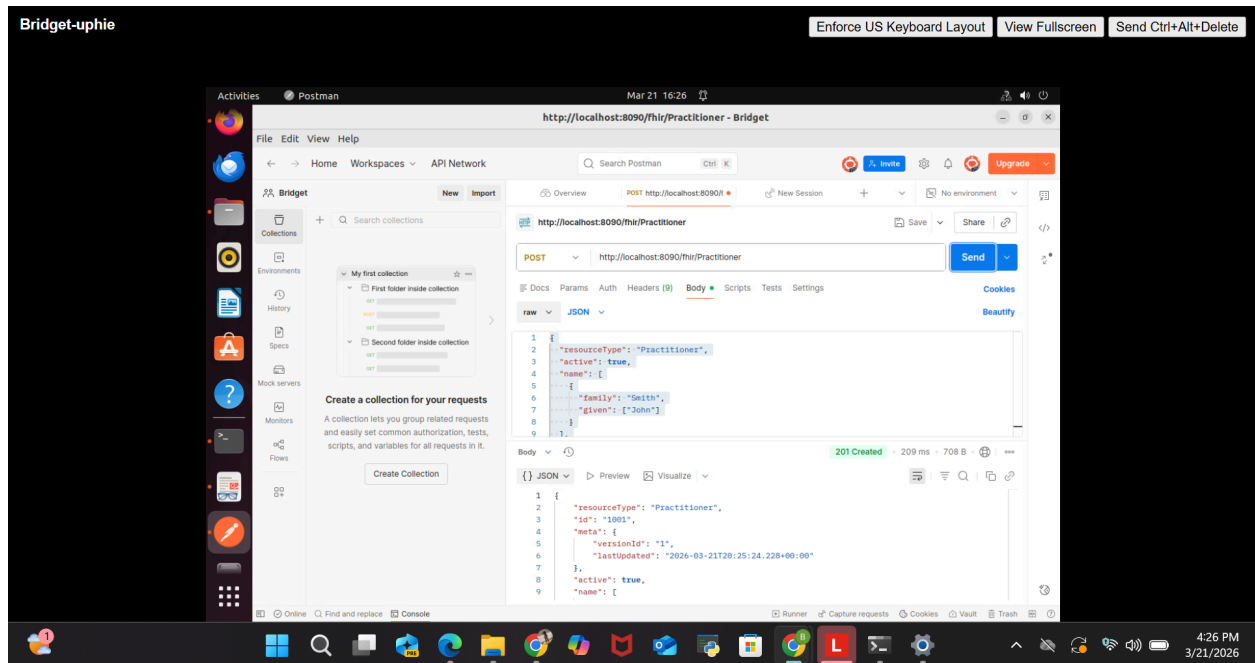
iii. Then, visit the official documentation to copy a sample JSON request body:

<https://docs.oracle.com/en/industries/health/millennium-platform-apis/mfrap/op-practitioner-post.html>

iv. After copying the JSON, paste it into the request body section in Postman and set the body type to raw and JSON.

v. Finally, click the Send button to execute the request.

vi. If the request is successful, an HTTP status code “201 Created” will be returned, along with a response containing the new resource ID, as shown in the screenshot below.



2. What is the process to manage authenFcaFon when using Postman with a HAPI FHIR server? (5

The first step is to open Postman and create or select a new HTTP request. Within the request interface, click on the Authorization tab, where different authentication methods are displayed.

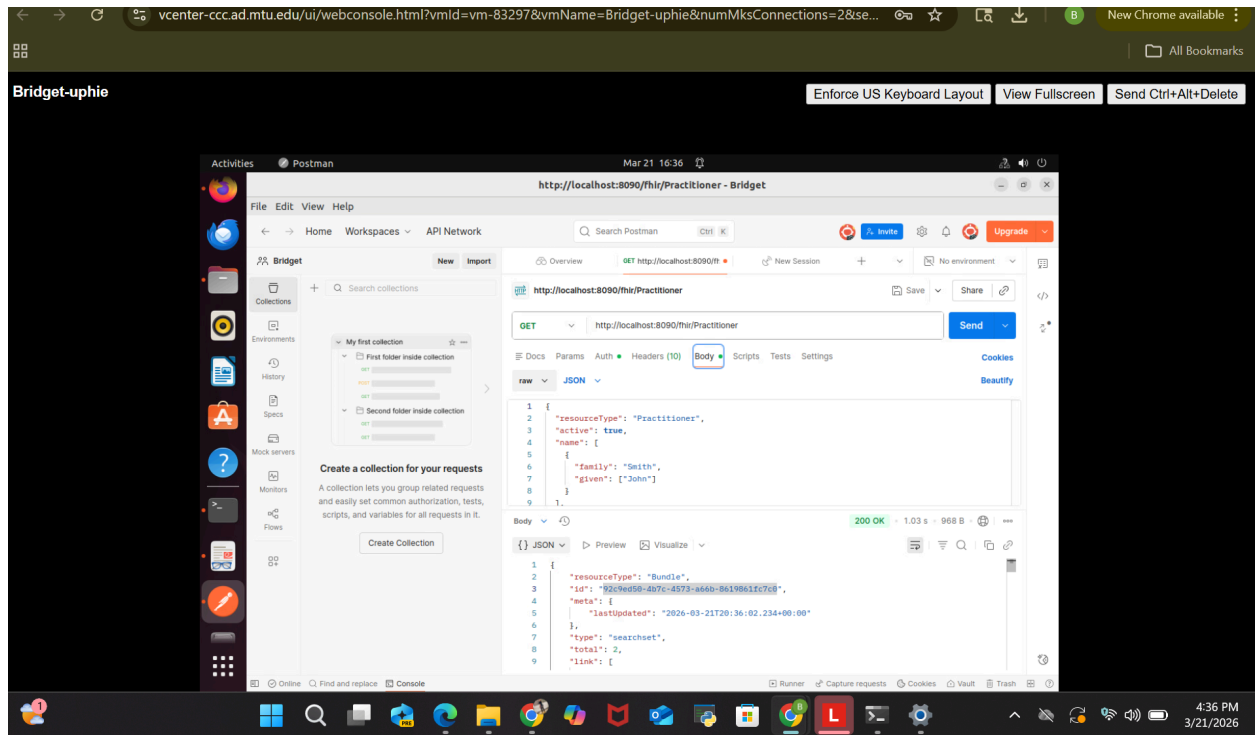
i. From the dropdown menu in the Authorization tab, choose the authentication method required by the HAPI FHIR server. Available options include Basic Auth, Bearer Token, and API Key, among others.

ii. In this case, select Basic Authorization (Basic Auth).

iii. After selecting Basic Auth, enter your username and password as the authentication credentials.

iv. Postman will then automatically generate and attach the appropriate Authorization headers to the HTTP request. This ensures that each request includes the correct credentials without needing manual header configuration.

v. Once authentication is successful and the request is valid, the server returns status codes such as **200 OK** (for GET, PUT, DELETE) or **201 Created** (for POST), along with a JSON response containing the resource data or confirmation message.



3. How can you handle error responses from the HAPI FHIR server when using Postman? (5 points)

i. First, understand common HTTP errors that may occur:

- 400 Bad Request – malformed input or missing parameters
- 404 Not Found – resource not found

- 422 Unprocessable Entity – validation errors
- 500 Internal Server Error – server-side issue

ii. When an error occurs, check the response body. The server usually returns an *OperationOutcome* with details such as severity, error code, and description.

iii. Use this information to correct the request.

iv. Finally, resend the request after fixing the error.